MATHEMATICS

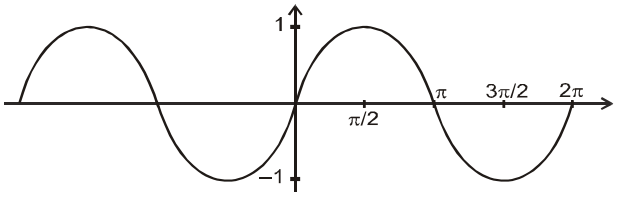
INVERSE TRIGONOMETRIC FUNCTIONS THEORY SHEET ENGLISH

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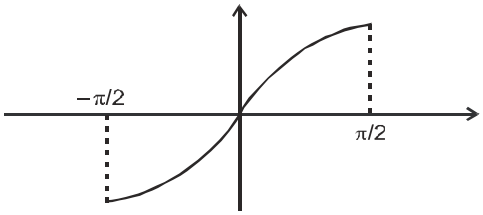
1.1 INTRODUCTION OF ITF

Let a function $f: A \to B$, $y: f(x)$ is such that function is one-one & onto then its inverse exists.If function is not bijective, then its inverse function cannot exist.

Consider a function $f: R \to R$, $y = \sin x$, this is neither one-one function nor ontofunction.



but we can make this function to behave like bijective function. To achieve this we have to restrict the domain and co-domain of function. Now let the domain of above functions is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and co-domain is $[-1, 1]$, then its graph becomes :



This shows that now y = sin x behaves like a one-one and onto function and we can calculate its inverse which is, $f: [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $y = \sin^{-1} x$.

Thus, $\sin^{-1} x$ is an angle, whose sine is x,

Note:

1. $\sin^{-1} x \neq \frac{1}{\sin x}$ 2. $\sin^{-1} x = \text{arc}(\sin x)$

1.2 DOMAIN AND RANGE OF 6 ITF

	***f* (x)**	**Domain**	**Range**
(1)	$\sin^{-1}x$	$\|x\| \le 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(2)	$\cos^{-1}x$	$\|x\| \le 1$	$[0, \pi]$
(3)	$\tan^{-1}x$	$x \in R$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(4)	$\cot^{-1}x$	$x \in R$	$(0, \pi)$
(5)	$\sec^{-1}x$	$\|x\| \ge 1$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
(6)	$\text{cosec}^{-1}x$	$\|x\| \ge 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Find Domain

1. $f(x) = \cos^{-1}(2x)$

Ans. $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

Sol. $2x \in [-1, 1]$

2. $f(x) = \tan^{-1}(\sqrt{x})$

Ans. $x \in [0, \infty)$

Sol. $x \geq 0$

3. $f(x) = \sin^{-1}(\ln x)$

Ans. $x \in \left[\frac{1}{e}, e\right]$

Sol. $\ln x \in [-1, 1]$

$x \in [e^{-1}, e]$

4. $f(x) = \sec^{-1}(e^x)$

Ans. $x \in [0, \infty)$

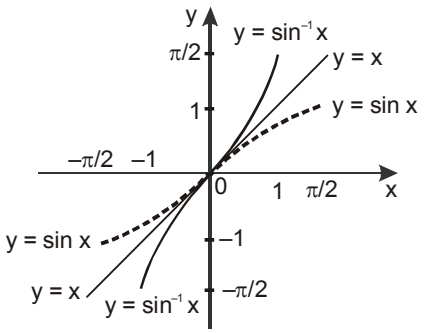
Sol. $e^x \in [1, \infty)$

$x \in [0, \infty)$

1.3 GRAPH OF 6 ITFs

1. $y = \sin^{-1} x$, $|x| \leq 1$, $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

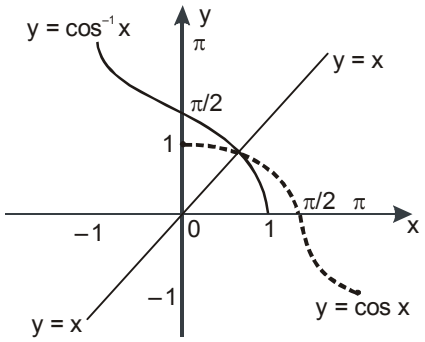
Domain	: [–1 , 1]
Range	: [– π/2 , π/2]
Nature	: one-one



- $\sin^{-1} x$ is an increasing function
- $\sin^{-1} x$ is an odd function
- $\sin^{-1} x$ is a bounded
- $\sin^{-1} x$ is aperiodic
- Concave up when $x > 0$, concave down when $x < 0$
- $x = 0$ is point of inflection
- Maximum value $= \frac{\pi}{2}$ when $x = 1$ and minimum value $= \frac{-\pi}{2}$ when $x = -1$

2. $y = \cos^{-1} x$, $|x| \le 1$, $y \in [0 , \pi]$

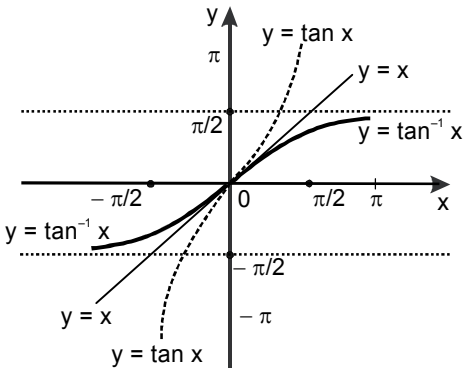
Domain	:	[–1 , 1]
Range	:	[0 , π]
Nature	:	one-one



- $\cos^{-1} x$ is a decreasing function
- $\cos^{-1} x$ is neither even nor odd function
- $\cos^{-1} x$ is bounded
- aperiodic
- Concave up when $x < 0$, concave down when $x > 0$
- $x = 0$ is point of inflection
- Maximum value = π when $x = -1$, minimum value = 0 when $x = 1$

3. $y = \tan^{-1} x$, $x \in R$, $y \in \left(-\frac{\pi}{2} , \frac{\pi}{2}\right)$

Domain	:	R
Range	:	(– π/2 , π/2)
Nature	:	one-one

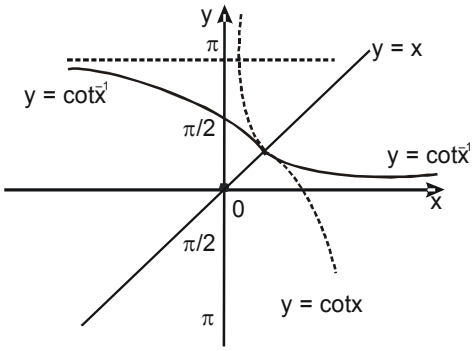


- $\tan^{-1} x$ is an increasing function
- $\tan^{-1} x$ is odd function
- $\tan^{-1} x$ is bounded
- aperiodic
- Concave up when $x < 0$, concave down when $x > 0$

- $x = 0$ is point of inflection
- Maximum value $\to \frac{\pi}{2}$ when $x \to \infty$, Minimum value $\to \frac{-\pi}{2}$ when $x \to -\infty$

4. $y = \cot^{-1} x\ ,\ x \in R\ ,\ y \in (0\ ,\ \pi)$

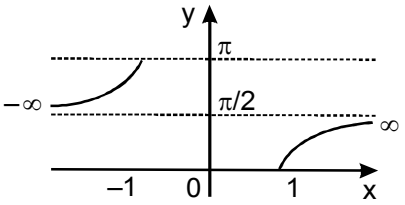
Domain	: R
Range	: $(0\ ,\ \pi)$
Nature	: one-one



- $\cot^{-1} x$ is a decreasing function in R
- $\cot^{-1} x$ is neither even nor odd function
- $\cot^{-1} x$ is bounded function
- aperiodic
- Concave up when $x > 0$, concave down when $x < 0$
- $x = 0$ is point of inflection
- Maximum value $\to \pi$ when $x \to -\infty$, Minimum value $\to 0$ when $x \to \infty$

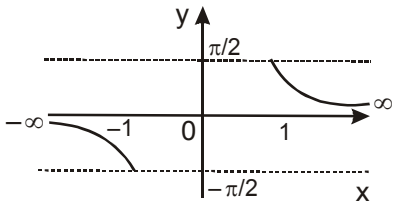
5. $y = \sec^{-1} x\ ,\ |x| \geq 1\ ,\ y \in \left[0\ ,\ \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}\ ,\ \pi\right]$

Domain	: $(-\infty\ ,\ -1] \cup [1\ ,\ \infty)$
Range	: $[0\ ,\ \pi] - \{\pi/2\}$
Nature	: one-one



6. $y = \text{cosec}^{-1} x\ ,\ |x| \geq 1\ ,\ y \in \left[-\frac{\pi}{2}\ ,\ 0\right) \cup \left(0\ ,\ \frac{\pi}{2}\right]$

Domain	: $(-\infty\ ,\ -1] \cup [1\ ,\ \infty)$
Range	: $[-\pi/2\ ,\ \pi/2] - \{0\}$
Nature	: one-one



Note :

1. Functions of first family are odd. $f(-x) = -f(x)$

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\sin^{-1}\left(\frac{1}{2}\right)$$

$\tan^{-1}(-2) = -\tan^{-1}(2)$

2. Functions of second family are NENO.

$f(x) + f(-x) = \pi \Rightarrow f(-x) = \pi - f(x)$

$$\cos^{-1}\left(\frac{-1}{4}\right) = \pi - \cos^{-1}\left(\frac{1}{4}\right)$$

$\cot^{-1}(-5) = \pi - \cot^{-1}(5)$

$\sec^{-1}(-4) = \pi - \sec^{-1}(4)$

3. $\sin^{-1}(x)$ and $\tan^{-1}(x)$ are increasing function

4. $\cos^{-1}(x)$ and $\cot^{-1}(x)$ are decreasing function

2.1 INEQUALITIES INVOLVING ITF

(i) $\sin^{-1}(x) > \sin^{-1}(y) \Rightarrow 1 \geq x > y \geq -1$

(ii) $\cos^{-1}(x) > \cos^{-1}(y) \Rightarrow -1 \leq x < y \leq 1$

(iii) $\tan^{-1}(x) > \tan^{-1}(y) \Rightarrow x > y$

(iv) $\cot^{-1}(x) > \cot^{-1}(y) \Rightarrow x < y$

Solve

1. $\sin^{-1}(x) > \sin^{-1}(2x-1)$

Ans. $x \in [0,1)$

Sol. $x > 2x - 1$ & $x \in [-1, 1]$ & $2x - 1 \in [-1, 1]$

$x \in [0, 1)$

2. $\cos^{-1}(x) > \cos^{-1}(x^2)$

Ans. $x \in [-1,0)$

Sol. $\cos^{-1}(x) > \cos^{-1}(x^2)$

$x < x^2$ & $x \in [-1, 1]$

$x(x-1) > 0$

$(-\infty, 0) \cup (1, \infty)$

$x \in [-1, 0)$

3. $\tan^{-1}(x+2) < \tan^{-1}(x^2)$

Ans. $x \in (-\infty, -1) \cup (2, \infty)$

Sol. $x^2 > x + 2$

$x^2 - x - 2 > 0$

$(x-2)(x+1) > 0$

4. $\sin^{-1}(x) > \frac{\pi}{4}$

Ans. $x \in \left(\frac{1}{\sqrt{2}}, 1\right]$

Sol. $x > \frac{1}{\sqrt{2}}$ & $x \in [-1, 1]$

$x \in \left(\frac{1}{\sqrt{2}}, 1\right]$

5. $\sin^{-1}(x) > \frac{\pi}{3}$

Ans. $x \in \left(\frac{\sqrt{3}}{2}, 1\right]$

Sol. $\frac{\sqrt{3}}{2} < x \le 1$

6. $\cos^{-1}(x) < \frac{2\pi}{3}$

Ans. $x \in \left(-\frac{1}{2}, 1\right]$

Sol. $-\frac{1}{2} < x \le 1$

7. $\cot^{-1}(x) < \frac{\pi}{6}$

Ans. $x \in (\sqrt{3}, \infty)$

Sol. $\sqrt{3} < x < \infty$

8. $\sin^{-1}(x) > \frac{2\pi}{3}$.

Ans. $x \in \phi$

Sol. $\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \not> \frac{2\pi}{3}$

$x \in \phi$

9. $\sin^{-1}(x) < \frac{2\pi}{3}$

Ans. $x \in [-1, 1]$

Sol. $\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] < \frac{2\pi}{3}$

$x \in [-1, 1]$

10. $\cot^{-1}(x) > -\frac{\pi}{3}$

Ans. $x \in R$

Sol. $\cot^{-1} x \in [0, \pi] > -\frac{\pi}{3}$

$x \in R$

2.2 DOMAIN BASED QUESTIONS

Find Domain of following functions

1. $f(x) = \sin^{-1}(2x + 3)$

Ans. $x \in [-2, -1]$

Sol. $2x + 3 \in [-1, 1]$

2. $f(x) = \sqrt{\cos^{-1}(2x+1)}$

Ans. $x \in [-1, 0]$

Sol. $\cos^{-1}(2x+1) \geq 0$ & $2x+1 \in [-1, 1]$

$2x+1 \leq 1$ $\quad x \in [-1, 0]$

$x \leq 0$

3. $f(x) = \sqrt{\sin^{-1}(2x) - \frac{\pi}{6}}$

Ans. $x \in \left[\frac{1}{4}, \frac{1}{2}\right]$

Sol. $\sin^{-1}(2x) \geq \frac{\pi}{6}$ & $2x \in [-1, 1]$

$2x \geq \frac{1}{2}$ $\quad x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

$x \geq \frac{1}{4}$

4. $f(x) = \ln(\cot^{-1}(x))$

Ans. $x \in R$

Sol. $\cot^{-1}x > 0$

5. $f(x) = \ln(\sin^{-1}(x))$

Ans. $x \in (0,1]$

Sol. $\sin^{-1}x > 0$ & $x \in [-1, 1]$

$x \in (0, 1]$

6. $f(x) = \sin^{-1}(x) + \sec^{-1}(x)$

Ans. $x \in \{-1,1\}$

Sol. $x \in [-1, 1]$ & $x \in (-\infty, -1] \cup [1, \infty)$

7. $f(x) = \cot^{-1}\sqrt{\sin^{-1}(x)}$

Ans. $x \in [0,1]$

Sol. $\sin^{-1}x \geq 0$

8. $f(x) = \log_{\cot^{-1}x} 2$.

Ans. $x \in R - \{\cot 1\}$

Sol. $\cot^{-1}x > 0$ & $\cot^{-1}x \neq 1$

2.3 RANGE BASED QUESTIONS

Find range of following functions

1. $f(x) = \sin^{-1}(e^x)$

Ans. $\left(0, \frac{\pi}{2}\right]$

Sol. $y = \sin^{-1}(0, 1]$

2. $f(x) = \sin^{-1}(2x + 3)$

Ans. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Sol. $y = \sin^{-1}[-1,1]$

3. $f(x) = \sqrt{\sin^{-1}(2x+3)}$

Ans. $\left[0, \sqrt{\frac{\pi}{2}}\right]$

Sol. $\sqrt{\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]}$

$\Rightarrow \sqrt{\left[0, \frac{\pi}{2}\right]}$

4. $f(x) = \sqrt{\cos^{-1}(2x) - \frac{\pi}{3}}$

Ans. $\left[0, \sqrt{\frac{2\pi}{3}}\right]$

Sol. $y = \sqrt{[0, \pi] - \frac{\pi}{3}}$

$= \sqrt{\left[-\frac{\pi}{3}, \frac{2\pi}{3}\right]} = \sqrt{\left[0, \frac{2\pi}{3}\right]}$

5. $f(x) = \ln(\cot^{-1} x)$

Ans. $(-\infty, \ln \pi)$

Sol. $y = \ell n(0, \pi)$

6. $f(x) = l\text{n}\,(\sin^{-1}x)$

Ans. $\left(-\infty, \ln\left(\frac{\pi}{2}\right)\right]$

Sol. $y = \ell n\left(0, \frac{\pi}{2}\right]$

7. $f(x) = \cot^{-1}(e^x)$

Ans. $\left(0, \frac{\pi}{2}\right)$

Sol. $y = \cot^{-1}(0, \infty)$

8. $f(x) = \cos^{-1}\{x\}$

Ans. $\left(0, \frac{\pi}{2}\right]$

Sol. $y = \cos^{-1}[0,1) = \left(0, \frac{\pi}{2}\right]$

9. $f(x) = \cot^{-1}\{x\}$

Ans. $\left(\frac{\pi}{4}, \frac{\pi}{2}\right]$

Sol. $y = \cot^{-1}[0, 1)$

10. $f(x) = \sin^{-1}[x]$

Ans. $\left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$

Sol. $y = \sin^{-1}\{-1,0,1\}$

11. $f(x) = \cos^{-1}[x]$

Ans. $\left\{0, \frac{\pi}{2}, \pi\right\}$

Sol. $y = \cos^{-1}\{-1,0,1\}$

12. $f(x) = \tan^{-1}\{x\}$

Ans. $\left[0, \frac{\pi}{4}\right)$

Sol. $\tan^{-1}[0, 1) = \left[0, \frac{\pi}{4}\right)$

13. $f(x) = \cot^{-1}(\ell n(x^2 + 2x + 2))$

Ans. $\left(0, \frac{\pi}{2}\right]$

Sol. $y = \cot^{-1}(\ell n((x + 1)^2 + 1))$

$y = \cot^{-1}(\ell n[1, \infty))$

$y = \cot^{-1}[0, \infty)$

14. $f(x) = \tan^{-1}(e^{2x} + e^x + 2)$

Ans. $\left(\tan^{-1}(2), \frac{\pi}{2}\right)$

Sol. $y = \tan^{-1}(2, \infty)$

15. $f(x) = \sin^{-1}(x) + \sec^{-1}(x)$

Ans. $\left\{\frac{\pi}{2}\right\}$

Sol. domain = $\{-1,1\}$

$$f(-1) = -\frac{\pi}{2} + \pi = \frac{\pi}{2}$$

$$f(1) = \frac{\pi}{2} + 0 = \frac{\pi}{2}$$

2.4 EVALUATION OF INVERSE TRIGONOMETRIC EXPRESSIONS

Find value of

1. $\cos(\tan^{-1} 2)$

Ans. $\frac{1}{\sqrt{5}}$

Sol. $\cos\left(\cos^{-1}\frac{1}{\sqrt{5}}\right)$

2. $\tan\left(\cos^{-1}\frac{5}{6}\right)$

Ans. $\frac{\sqrt{11}}{5}$

Sol. $\tan\left(\tan^{-1}\frac{\sqrt{11}}{5}\right)$

3. $\cot\left(\tan^{-1}\frac{4}{3}\right)$

Ans. $\frac{3}{4}$

Sol. $\cot\left(\cot^{-1}\frac{4}{3}\right)$

4. $\sin\left(2\sin^{-1}\frac{4}{5}\right)$

Ans. $\frac{24}{25}$

Sol. $2\sin\left(\sin^{-1}\frac{4}{5}\right)\cos\left(\sin^{-1}\frac{4}{5}\right)$

$2\times\frac{4}{5}\times\frac{3}{5}=\frac{24}{25}$

5. $\cos(2\tan^{-1}(3))$

Ans. $-\frac{4}{5}$

Sol. $\frac{1-\tan^2\left(\tan^{-1}3\right)}{1+\tan^2\left(\tan^{-1}3\right)}=\frac{1-9}{1+9}=-\frac{8}{10}$

6. $\sin(2\tan^{-1}2)$

Ans. $\frac{4}{5}$

Sol. $\dfrac{2\tan\left(\tan^{-1}2\right)}{1+\left(\tan\left(\tan^{-1}2\right)\right)^2}=\dfrac{4}{5}$

7. $\sin\left(2\tan^{-1}\left(\dfrac{1}{2}\right)\right)$

Ans. $\dfrac{4}{5}$

Sol. $\dfrac{2\left(\tan\left(\tan^{-1}\dfrac{1}{2}\right)\right)}{1+\left(\tan\left(\tan^{-1}\dfrac{1}{2}\right)\right)^2}=\dfrac{4}{5}$

8. $\sin\left(\sin^{-1}\left(\dfrac{3}{5}\right)-\cos^{-1}\left(\dfrac{3}{5}\right)\right)$

Ans. $-\dfrac{7}{25}$

Sol. $\sin\left(\sin^{-1}\dfrac{3}{5}\right)\cos\left(\cos^{-1}\dfrac{3}{5}\right)-\cos\left(\sin^{-1}\dfrac{3}{5}\right)\cdot\sin\left(\cos^{-1}\dfrac{3}{5}\right)$

$=\dfrac{3}{5}\times\dfrac{3}{5}-\dfrac{4}{5}\times\dfrac{4}{5}=\dfrac{9-16}{25}$

9. $\tan\left(\dfrac{1}{2}\cos^{-1}\left(\dfrac{\sqrt{5}}{3}\right)\right)$

Ans. $\dfrac{3-\sqrt{5}}{2}$

Sol. let $y=\tan\left(\dfrac{1}{2}\cos^{-1}\dfrac{\sqrt{5}}{3}\right)=\tan\left(\dfrac{\theta}{2}\right)$

let $\cos^{-1}\dfrac{\sqrt{5}}{3}=\theta$

$$\cos\theta = \frac{\sqrt{5}}{3},\ \sin\theta = \frac{2}{3}$$

$$\tan\frac{\theta}{2} = \frac{1-\cos\theta}{\sin\theta} = \frac{1-\frac{\sqrt{5}}{3}}{\frac{2}{3}} = \frac{3-\sqrt{5}}{2}$$

10. $\sin\left(\frac{1}{2}\sin^{-1}\left(\frac{5}{13}\right)\right)$

Ans. $\frac{1}{\sqrt{26}}$

Sol. Let $\sin^{-1}\frac{5}{13} = \theta,\ \sin\theta = \frac{5}{13}$

$$\sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\frac{12}{13}}{2}} = \frac{1}{\sqrt{26}}$$

11. $\tan\left(\frac{1}{2}\tan^{-1}\left(\frac{5}{12}\right)\right)$

Ans. $\frac{1}{5}$

Sol. $\tan\left(\frac{1}{2}\tan^{-1}\frac{5}{12}\right)$

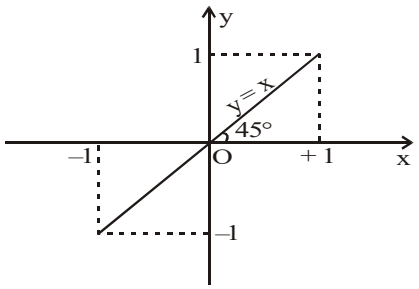
Let $\tan\theta = \frac{5}{12}, \sin\theta = \frac{5}{13}, \cos\theta = \frac{12}{13}$

$$\tan\frac{\theta}{2} = \frac{1-\cos\theta}{\sin\theta} = \frac{1-\frac{12}{13}}{\frac{5}{13}} = \frac{1}{5}$$

3.1 PROPERTY 1 OF ITF

(i) $y = \sin(\sin^{-1}x) = x$

$x \in [-1,1], y \in [-1, 1]$

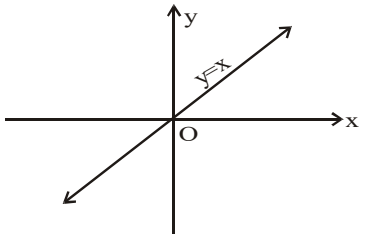


(ii) $y = \cos(\cos^{-1} x) = x$

$x \in [-1,1], y \in [-1,1]$

(iii) $y = \tan(\tan^{-1} x) = x$

$x \in R, y \in R$

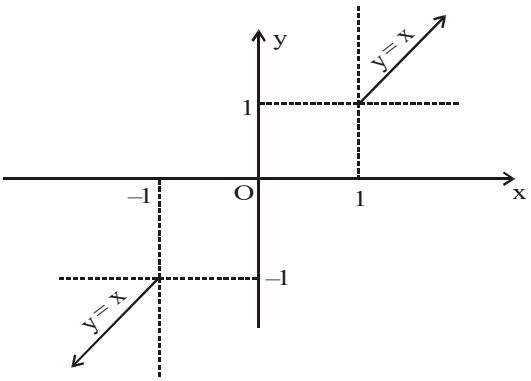


(iv) $y = \cot(\cot^{-1} x) = x,$

$x \in R; y \in R$

(v) $y = \text{cosec}(\text{cosec}^{-1} x) = x,$

$|x| \geq 1, |y| \geq 1$



(vi) $y = \sec(\sec^{-1} x) = x$

$|x| \geq 1 ; |y| \geq 1$

3.2 PROPERTY 2 OF ITF

1. $y = \sin^{-1}(\sin x),$

$x \in R,\ y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

periodic with period 2π; odd function.

$\sin y = \sin x$

$\sin x = \sin y$

$x = n\pi + (-1)^n y$

$x = y \Rightarrow y = x \qquad x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$x = \pi - y \Rightarrow y = \pi - x \; x \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

$x = 2\pi + y \Rightarrow y = x - 2\pi \qquad x \in \left[\frac{3\pi}{2}, \frac{5\pi}{2}\right]$

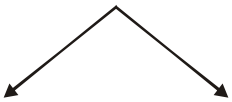
$x = 3\pi - y \Rightarrow y = 3\pi - x \qquad x \in \left[\frac{5\pi}{2}, \frac{7\pi}{2}\right]$

$x = 4\pi + y \Rightarrow y = x - 4\pi \qquad x \in \left[\frac{7\pi}{2}, \frac{9\pi}{2}\right]$

$\sin^{-1}\sin(x)$

↓

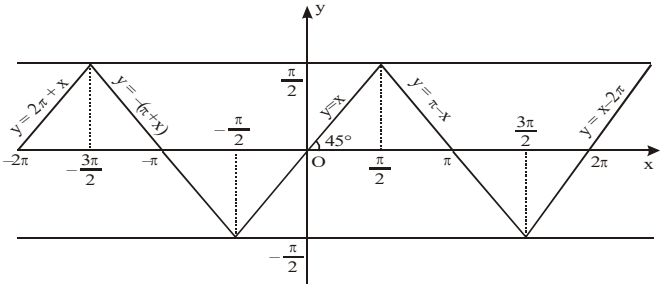
round off to nearest $n\pi$



n = even $\qquad$ n = odd

$\sin^{-1}\sin(x) = x - \text{even}\ \pi \qquad \sin^{-1}\sin(x) = \text{odd}\ \pi - x$

Graph:



2. $y = \text{cosec}^{-1}(\text{cosec}\, x)$,

$x \in R - \{n\pi, n \in I\} \qquad y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right],$

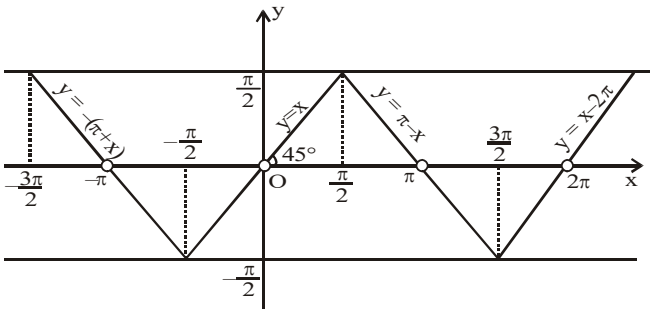
periodic with period 2π; $\qquad$ odd function.

$\text{cosec}(y) = \text{cosec}(x)$

$\sin(y) = \sin(x)$

same as that of $\sin^{-1}(\sin x)$

Graph:



3. $y = \cos^{-1}(\cos x)$,

$x \in R$, $y \in [0,\pi]$,

periodic with period 2π; even function.

$\cos y = \cos x$

$\cos x = \cos y$

$x = 2n\pi \pm y$

$x = +y \quad \Rightarrow \quad y = +x \; x \in [0, \pi]$

$x = 2\pi - y \quad \Rightarrow \quad y = 2\pi - x \quad x \in [\pi, 2\pi]$

$x = 2\pi + y \quad \Rightarrow \quad y = x - 2\pi \quad x \in [2\pi, 3\pi]$

$x = 4\pi - y \quad \Rightarrow \quad y = 4\pi - x \quad x \in [3\pi, 4\pi]$

$x = 4\pi + y \quad \Rightarrow \quad y = x - 4\pi \quad x \in [4\pi, 5\pi]$

$\cos^{-1}\cos(x)$

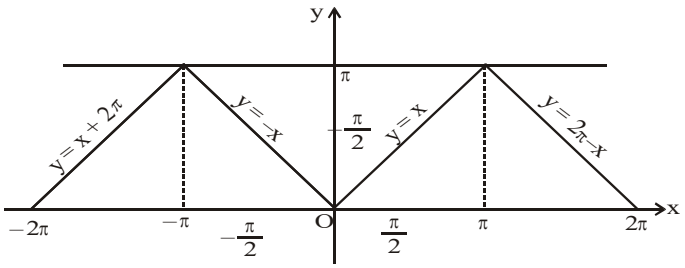
↓

round off to nearest even π

if x > even π: $\cos^{-1}\cos(x) = x - \text{even } \pi$

if x < even π: $\cos^{-1}\cos(x) = \text{even } \pi - x$

Graph:



4. $y = \sec^{-1}(\sec x)$,

$x \in R - \left\{(2n-1)\frac{\pi}{2} n \in I\right\}, \quad y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

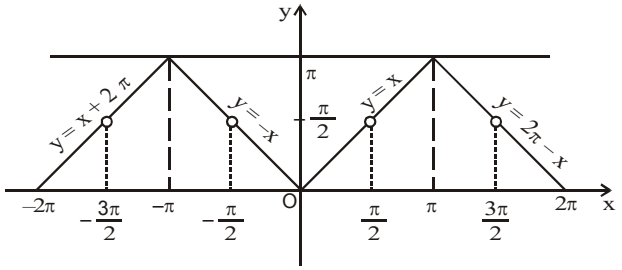
periodic with period 2π ; even function.

$\sec(y) = \sec(x)$

$\cos(y) = \cos(x)$

same as that of $\cos^{-1}(\cos x)$

Graph:



5. $y = \tan^{-1}(\tan x)$

$x \in R - \left\{(2n-1)\frac{\pi}{2}, n \in I\right\}; \quad y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$

periodic with period π

odd function.

$\tan y = \tan x$

$y = x \qquad x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

$y = x - \pi \qquad x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

$y = x - 2\pi \qquad x \in \left(\frac{3\pi}{2}, \frac{4\pi}{2}\right)$

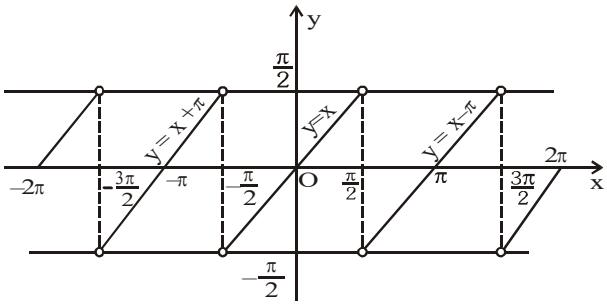
$y = x - 3\pi \qquad x \in \left(\frac{5\pi}{2}, \frac{7\pi}{2}\right)$

$y = x - 4\pi \qquad x \in \left(\frac{7\pi}{2}, \frac{5\pi}{2}\right)$

$\tan^{-1}\tan(x) \rightarrow$ Round off to nearest $n\pi$

$\tan^{-1}\tan(x) = x - n\pi$

Graph:



6. $y = \cot^{-1}(\cot x)$,

$x \in R - \{n\pi, n \in I\}$, $y \in (0, \pi)$,

periodic with period π ; neither even nor odd function.

$\cot y = \cot x$

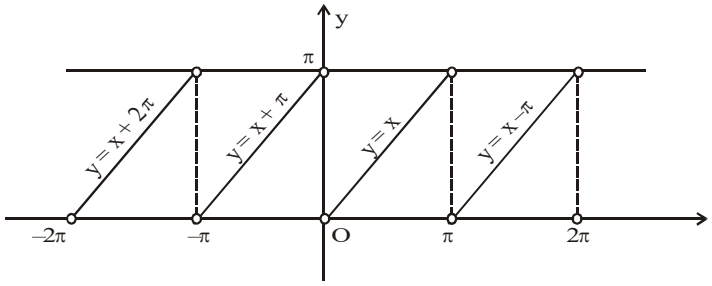
$\tan y = \tan x$

$y = x$	$x \in (0, \pi)$
$y = x - \pi$	$x \in (\pi, 2\pi)$
$y = x - 2\pi$	$x \in (2\pi, 3\pi)$
$y = x - 3\pi$	$x \in (3\pi, 4\pi)$
$y = x - 4\pi$	$x \in (4\pi, 5\pi)$

$\cot^{-1} \cot(x) \rightarrow$ Change to integral part of π

$\cot^{-1} \cot(x) = x - [\,]\pi$

Graph



Define

1. $y = \sin^{-1} \sin(x)$; $x \in [-\pi, 2\pi]$

Ans. $y = \begin{cases} -\pi - x & -\pi \le x \le -\frac{\pi}{2} \\ x & -\frac{\pi}{2} \le x \le \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} \le x \le \frac{3\pi}{2} \\ x - 2\pi & \frac{3\pi}{2} \le x \le 2\pi \end{cases}$

2. $y = \cos^{-1} \cos(2x);\ x \in [0,\ \pi]$

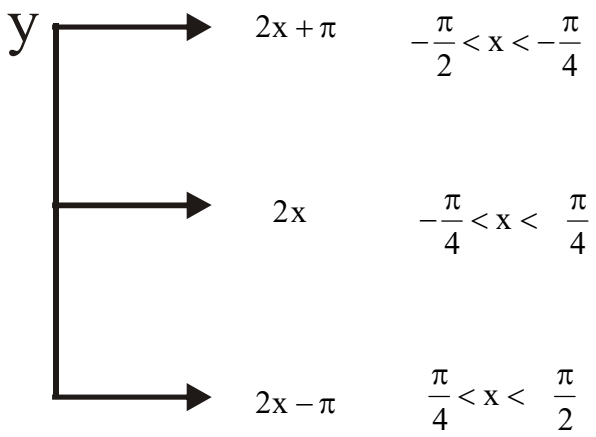
Ans. $y \begin{cases} 2x & 0 \le x \le \frac{\pi}{2} \\ 2\pi - 2x & \frac{\pi}{2} \le x \le \pi \end{cases}$

3. $y = \tan^{-1} \tan(x);\ x \in [0,\ 2\pi]$

Ans. $y \begin{cases} x & 0 \le x < \frac{\pi}{2} \\ x - \pi & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ x - 2\pi & \frac{3\pi}{2} < x \le 2\pi \end{cases}$

4. $y = \tan^{-1}(\tan 2x);\ x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

Ans.



3.3 SIMPLIFICATION BY TRIGONOMETRIC SUBSTITUTION

put $x = \sin\theta$ or $\theta = \sin^{-1}x$ $\quad \theta \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

put $x = \cos\theta$ or $\theta = \cos^{-1}x$ $\quad \theta \in [0, \pi]$

put $x = \tan\theta$ or $\theta = \tan^{-1}x$ $\quad \theta \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

Simplify

1. $y = \cos^{-1}(2x^2 - 1)$

Ans $y = \cos^{-1}(2x^2 - 1) = \begin{cases} 2\cos^{-1}x & 0 \le x \le 1 \\ 2\pi - 2\cos^{-1}x & -1 \le x \le 0 \end{cases}$

2. $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

Ans $y = f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & \text{if} \quad |x| \le 1 \\ \pi - 2\tan^{-1}x & \text{if} \quad x > 1 \\ -(\pi + 2\tan^{-1}x) & \text{if} \quad x < -1 \end{cases}$

3. $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Ans. $y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & \text{if} \quad x \ge 0 \\ -2\tan^{-1}x & \text{if} \quad x < 0 \end{cases}$

4. $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

Ans $y = f(x) = \tan^{-1}\frac{2x}{1-x^2} = \begin{cases} 2\tan^{-1}x & \text{if} \quad |x|<1 \\ \pi + 2\tan^{-1}x & \text{if} \quad x<-1 \\ -(\pi - 2\tan^{-1}x) & \text{if} \quad x>1 \end{cases}$

5. $y = \sin^{-1}(3x - 4x^3)$

Ans $y = f(x) = \sin^{-1}(3x - 4x^3) = \begin{cases} -(\pi + 3\sin^{-1}x) & \text{if} \quad -1 \le x \le -\frac{1}{2} \\ 3\sin^{-1}x & \text{if} \quad -\frac{1}{2} \le x \le \frac{1}{2} \\ \pi - 3\sin^{-1}x & \text{if} \quad \frac{1}{2} \le x \le 1 \end{cases}$

6. $y = \cos^{-1}(4x^3 - 3x)$

Ans $y = \cos^{-1}(4x^3 - 3x) = \begin{cases} 3\cos^{-1}x - 2\pi & \text{if} \quad -1 \le x \le -\frac{1}{2} \\ 2\pi - 3\cos^{-1}x & \text{if} \quad -\frac{1}{2} \le x \le \frac{1}{2} \\ 3\cos^{-1}x & \text{if} \quad \frac{1}{2} \le x \le 1 \end{cases}$

7. $y = \tan^{-1}\left(\frac{3x - x^3}{1-3x^2}\right)$

Ans $\tan^{-1}\frac{3x - x^3}{1-3x^2} = \begin{cases} 3\tan^{-1}x & \text{if} \quad -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}} \\ -\pi + 3\tan^{-1}x & \text{if} \quad x > \frac{1}{\sqrt{3}} \\ \pi + 3\tan^{-1}x & \text{if} \quad x < -\frac{1}{\sqrt{3}} \end{cases}$

3.4 PROPERTY 3 OF ITF(RELATION BETWEEN f (x) AND f (-x))

(i) $\sin^{-1}(-x) = -\sin^{-1} x$, $-1 \le x \le 1$

$\tan^{-1}(-x) = -\tan^{-1} x$, $x \in R$

$\text{cosec}^{-1}(-x) = -\text{cosec}^{-1} x$, $|x| \ge 1$

(ii) $\cot^{-1}(-x) = \pi - \cot^{-1} x$, $x \in R$

$\cos^{-1}(-x) = \pi - \cos^{-1} x$, $-1 \le x \le 1$

$\sec^{-1}(-x) = \pi - \sec^{-1} x$, $|x| \ge 1$

3.5 PROPERTY 4 OF ITF

(i) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ $-1 \le x \le 1$

(ii) $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ $x \in R$

(iii) $\text{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}$ $|x| \ge 1$

3.6 PROPERTY 5 OF ITF

(i) $\text{cosec}^{-1} x = \sin^{-1}\frac{1}{x}$; $|x| \ge 1$

(ii) $\sec^{-1} x = \cos^{-1}\frac{1}{x}$; $|x| \ge 1$

(iii) $$\cot^{-1} x = \begin{cases} \tan^{-1}\frac{1}{x} & ; \quad x > 0 \\ \pi + \tan^{-1}\frac{1}{x} & ; \quad x < 0 \end{cases}$$

Q.1 Solve: $4\sin^{-1}(x) + \cos^{-1}(x) = \frac{2\pi}{3}$.

Ans. $x = \sin\left(\frac{\pi}{18}\right)$

Sol. $3\sin^{-1} x + \frac{\pi}{2} = \frac{2\pi}{3}$

$3\sin^{-1} x = \frac{\pi}{6}$

Q.2 Solve: $5\tan^{-1} x + 3\cot^{-1}(x) = \frac{3\pi}{4}$.

Ans. $x = -\tan\frac{3\pi}{8}$

Sol. $2\tan^{-1}x + \frac{3\pi}{2} = \frac{3\pi}{4}$

$2\tan^{-1}x = -\frac{3\pi}{4}$

Q.3 Solve: $\sin^{-1}x > \cos^{-1}(x)$.

Ans. $x \in \left(\frac{1}{\sqrt{2}}, 1\right]$

Sol. $\sin^{-1}x > \frac{\pi}{2} - \sin^{-1}x$

$\sin^{-1}x > \frac{\pi}{4}$

Q.4 Prove that $\sin(\cos^{-1}x) = \cos(\sin^{-1}x)$

Sol. LHS $\sin\left(\frac{\pi}{2} - \sin^{-1}x\right)$

$= \cos(\sin^{-1}x) = \text{RHS}$

Q.5 $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ find range.

Ans. $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

Sol. domain $= [-1,1]$

$f(x) = \frac{\pi}{2} + \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

$= \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

Q.6 $f(x) = (\sin^{-1}x)^2 + (\cos^{-1}x)^2$ find range.

Ans. $\left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]$

Sol. Let $\cos^{-1}x = a \in [0, \pi]$

$$f(x)=\left(\frac{\pi}{2}\right)^2-2\left(\frac{\pi}{2}-a\right)a$$

$$=2a^2-\pi a+\frac{\pi^2}{4}$$

$$\min=\frac{\pi^2}{8}$$

$$\max=\frac{5\pi^2}{4}$$

Q.7 $f(x)=(\sin^{-1}x)^3+(\cos^{-1}x)^3$ find range.

Ans. $\left[\frac{\pi^3}{32},\frac{7\pi^3}{8}\right]$

Sol. $f(x)=\underbrace{\left(\sin^{-1}x\right)^3}_{a}+\underbrace{\left(\cos^{-1}x\right)^3}_{b}$

$y=a^3+b^3$

$y=(a+b)^3-3ab(a+b)$

$$=\left(\frac{\pi}{2}\right)^3-3a\left(\frac{\pi}{2}-a\right)\frac{\pi}{2}$$

$$=\frac{\pi^3}{8}-\frac{3\pi^2}{4}a+\frac{3\pi}{2}a^2$$

$$=\frac{3\pi}{2}a^2-\frac{3\pi^2}{4}a+\frac{\pi^3}{8},\ a\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$$

$$(f)_{min}=\frac{\pi^3}{32}$$

$$(f)_{max}=\frac{7\pi^3}{8}$$

Q.8 Prove that $\cos\left(\cot^{-1}\left(\sec\left(\sin^{-1}a\right)\right)\right)=\frac{1}{\sqrt{2-a^2}}$

Sol. $\cos\left(\cot^{-1}\left(\sec\left(\sec^{-1}\frac{1}{\sqrt{1-a^2}}\right)\right)\right)$

$\cos\left(\cot^{-1}\frac{1}{\sqrt{1-a^2}}\right)$

$\cos\left(\cos^{-1}\frac{1}{\sqrt{2-a^2}}\right)$

Q.9 Find value of $\sin^{-1}\left(\cos\left(2\cot^{-1}\left(\sqrt{2}-1\right)\right)\right)$

Ans. $-\frac{\pi}{4}$

Sol. $\sin^{-1}\left(\cos 2\times\frac{3\pi}{8}\right)$

$\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)=-\frac{\pi}{4}$

3.7 ADDITION AND SUBTRACTION OF ANGLES

(a) $$\tan^{-1}x+\tan^{-1}y=\begin{cases}\tan^{-1}\frac{x+y}{1-xy} & \text{where } x>0, y>0 \,\&\, xy<1\\ \pi+\tan^{-1}\frac{x+y}{1-xy} & \text{where } x>0, y>0 \,\&\, xy>1\\ \frac{\pi}{2} & \text{where } x>0, y>0 \,\&\, xy=1\end{cases}$$

(b) $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\frac{x-y}{1+xy}$ where $x>0, y>0$

Find value of

(i) $\tan^{-1}(3)-\tan^{-1}\left(\frac{1}{2}\right)$.

Ans. $\frac{\pi}{4}$

Sol. $\tan^{-1}\left(\dfrac{3-\dfrac{1}{2}}{1+\dfrac{3}{2}}\right)=\tan^{-1}\left(\dfrac{\dfrac{5}{2}}{\dfrac{5}{2}}\right)=\tan^{-1}(1)$

(ii) $2\tan^{-1}\left(\dfrac{3}{4}\right)$

Ans. $\tan^{-1}\left(\dfrac{24}{7}\right)$

Sol. $\tan^{-1}\dfrac{3}{4}+\tan^{-1}\dfrac{3}{4}$

$$\tan^{-1}\left(\frac{\frac{3}{4}+\frac{3}{4}}{1-\frac{3}{4}\times\frac{3}{4}}\right)=\tan^{-1}\left(\frac{24}{7}\right)$$

(iii) $2\tan^{-1}\left(\dfrac{4}{3}\right)$

Ans. $\pi-\tan^{-1}\left(\dfrac{24}{7}\right)$

Sol. $\tan^{-1}\dfrac{4}{3}+\tan^{-1}\dfrac{4}{3}$

$$\pi+\tan^{-1}\left(\frac{\frac{4}{3}+\frac{4}{3}}{1-\frac{4}{3}\times\frac{4}{3}}\right)$$

$$\pi-\tan^{-1}\frac{24}{7}$$

(iv) $\tan^{-1}(5)-\tan^{-1}(3)+\tan^{-1}\left(\dfrac{7}{9}\right)$

Ans. $\frac{\pi}{4}$

Sol. $\tan^{-1}5-\tan^{-1}3+\tan^{-1}\frac{7}{9}$

$\tan^{-1}\left(\frac{5-3}{1+5\times3}\right)+\tan^{-1}\frac{7}{9}$

$\tan^{-1}\left(\frac{1}{8}\right)+\tan^{-1}\frac{7}{9}$

$\tan^{-1}\left(\frac{\frac{1}{8}+\frac{7}{9}}{1-\frac{1}{8}\times\frac{7}{9}}\right)=\tan^{-1}(1)$

(v) $\sin^{-1}\left(\frac{4}{\sqrt{17}}\right)+\cot^{-1}\left(\frac{5}{2}\right)$

Ans. $\pi-\tan^{-1}\left(\frac{22}{3}\right)$

Sol. $\sin^{-1}\left(\frac{4}{\sqrt{17}}\right)+\cot^{-1}\frac{5}{2}$

$\tan^{-1}\left(\frac{4}{1}\right)+\tan^{-1}\left(\frac{2}{5}\right)$

$\pi+\tan^{-1}\left(\frac{\frac{4}{1}+\frac{2}{5}}{1-\frac{8}{5}}\right)$

$\pi-\tan^{-1}\left(\frac{22}{3}\right)$

(vi) $\sin^{-1}\left(\frac{3}{\sqrt{73}}\right)+\cos^{-1}\left(\frac{11}{\sqrt{146}}\right)+\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Ans. $\frac{7\pi}{12}$

Sol. $\sin^{-1}\left(\frac{3}{\sqrt{73}}\right)+\cos^{-1}\left(\frac{11}{\sqrt{146}}\right)+\cot^{-1}\left(\sqrt{3}\right)$

$\tan^{-1}\frac{3}{8}+\tan^{-1}\frac{5}{11}+\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$\tan^{-1}\left(\frac{\frac{3}{8}+\frac{5}{11}}{1-\frac{3}{8}\times\frac{5}{11}}\right)+\tan^{-1}\frac{1}{\sqrt{3}}$

$\tan^{-1}1+\tan^{-1}\frac{1}{\sqrt{3}}=\frac{\pi}{4}+\frac{\pi}{6}=\frac{5\pi}{12}$

(vii) $\tan^{-1}(4)+\frac{1}{2}\tan^{-1}\left(\frac{5}{12}\right)=\tan^{-1}(x)$, find x.

Ans. 21

Sol. $\tan^{-1}(4)+\frac{1}{2}\tan^{-1}\frac{5}{12}$

Let $\frac{1}{2}\tan^{-1}\frac{5}{12}=\tan^{-1}(x)$

$\Rightarrow x=\tan\left(\frac{1}{2}\tan^{-1}\frac{5}{12}\right)$

Let $\tan^{-1}\frac{5}{12}=\theta$

$\tan\theta=\frac{5}{12},\sin\theta=\frac{5}{13}$

$\tan\frac{\theta}{2}=\frac{1-\frac{12}{13}}{\frac{5}{13}}=\frac{1}{5}$

$$\frac{1}{2}\tan^{-1}\frac{5}{12}=\tan^{-1}\frac{1}{5}$$

$$\tan^{-1}(4)+\tan^{-1}\left(\frac{1}{5}\right)=\tan^{-1}(x)$$

$$\tan^{-1}\left(\frac{4+\frac{1}{5}}{1-\frac{4}{5}}\right)=\tan^{-1}(x)$$

$$x=21$$

(viii) $\tan^{-1}\left(\frac{\sqrt{3}-\sqrt{2}}{1+\sqrt{6}}\right)+\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Ans. $\frac{\pi}{3}$

Sol. $\tan^{-1}\sqrt{3}-\tan^{-1}\sqrt{2}+\tan^{-1}\sqrt{2}$

$$=\frac{\pi}{3}$$

4. ITF EQUATIONS

Q.1 $2\cot^{-1}(2)-\cos^{-1}\left(\frac{4}{5}\right)=\text{cosec}^{-1}(x)$

Ans. $\frac{25}{7}$

Sol. $2\tan^{-1}\frac{1}{2}-\tan^{-1}\left(\frac{3}{4}\right)=\cos ec^{-1}x$

$$\tan^{-1}\left(\frac{\frac{1}{2}+\frac{1}{2}}{1-\frac{1}{4}}\right)-\tan^{-1}\left(\frac{3}{4}\right)=\cos ec^{-1}x$$

$$\tan^{-1}\left(\frac{4}{3}\right)-\tan^{-1}\left(\frac{3}{4}\right)=\cos ec^{-1}x$$

$$\tan^{-1}\left(\frac{\frac{4}{3}-\frac{3}{4}}{1+1}\right)=\cos ec^{-1}x$$

$$\tan^{-1}\left(\frac{7}{24}\right)=\cos ec^{-1}x$$

$$x=\frac{25}{7}$$

Q.2 $2\cot^{-1}(2)+\cos^{-1}\left(\frac{3}{5}\right)=\text{cosec}^{-1}(x)$

Ans. $x\in\phi$

Sol. $2\tan^{-1}\frac{1}{2}+\tan^{-1}\left(\frac{4}{3}\right)=\cos ec^{-1}x$

$$\tan^{-1}\left(\frac{4}{3}\right)+\tan^{-1}\frac{4}{3}=\cos ec^{-1}x$$

$$\pi+\tan^{-1}\left(\frac{\frac{8}{3}}{1-\frac{16}{3}}\right)=\cos ec^{-1}x$$

$$\underbrace{\pi-\tan^{-1}\left(\frac{24}{7}\right)}_{\text{OBTUSE}}=\underbrace{\cos ec^{-1}x}_{\max=\frac{\pi}{2}}$$

$x\in\phi$

Q.3 $\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}(x)=\frac{\pi}{4}$

Ans. $\frac{1}{3}$

Sol. $\tan^{-1}x=\frac{\pi}{2}-\tan^{-1}\frac{1}{2}$

$$\tan^{-1} x = \tan^{-1}\left(\frac{1-\frac{1}{2}}{1+\frac{1}{2}}\right)$$

$$\tan^{-1} x = \tan^{-1}\frac{1}{3}$$

Q.4 $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$

Ans. $\frac{1}{6}$

Sol. Let $\tan^{-1}(2x) = A$, $\tan^{-1}(3x) = B$

$$\tan(A + B) = \tan\frac{\pi}{4}$$

$$\frac{2x+3x}{1-6x^2} = 1$$

$$6x^2 + 5x - 1 = 0$$

$$x = \frac{1}{6}, -1 \text{ (Rejected)}$$

Q.5 $\tan^{-1}\left(\frac{x+1}{x-1}\right) + \tan^{-1}\left(\frac{x-1}{x}\right) = \tan^{-1}(-7)$

Ans. $x \in \phi$

Sol: Taking the tangents of both sides of the equation, we have

$$\frac{\tan\left[\tan^{-1}\frac{x+1}{x-1}\right] + \tan\left[\tan^{-1}\frac{x-1}{x}\right]}{1 - \tan\left[\tan^{-1}\frac{x+1}{x-1}\right]\tan\left[\tan^{-1}\frac{x-1}{x}\right]} = \tan\{\tan^{-1}(-7)\} = -7$$

i.e., $$\frac{\frac{x+1}{x-1} + \frac{x-1}{x}}{1 - \frac{x+1}{x-1}\frac{x-1}{x}} = -7$$

i.e., $$\frac{2x^2 - x + 1}{1 - x} = -7$$

so that $x = 2$.

This value makes the left hand side of the given equation positive, so that there is no value of x strictly satisfying the given equation.

The value $x = 2$ is a solution of the equation

$$\tan^{-1}\frac{x+1}{x-1} + \tan^{-1}\frac{x-1}{x} = \pi + \tan^{-1}(-7)$$

Q.6 $\sin^{-1}(x) = 2\tan^{-1}(x)$

Ans. $\{0, 1, -1\}$

Sol. $x = \sin(2\tan^{-1}x)$

$$x = \frac{2x}{1+x^2}$$

$x = 0$ or $1 + x^2 = 2$

$x = \pm 1$

Q.7 $\cos^{-1}(x) - \sin^{-1}(x) = \cos^{-1}(\sqrt{3}\,x)$

Ans. $\left\{0, \frac{1}{2}, \frac{-1}{2}\right\}$

Sol. $\underbrace{\cos^{-1}x}_{A} - \underbrace{\sin^{-1}x}_{B} = \cos^{-1}\left(x\sqrt{3}\right)$

$$\cos(A - B) = x\sqrt{3}$$

$$x\sqrt{1-x^2} + \sqrt{1-x^2}\,x = x\sqrt{3}$$

$$x = 0 \text{ or } 1 - x^2 = \frac{3}{4}$$

$$x = \pm\frac{1}{2}$$

Q.8 $\sin[2\cos^{-1}\{\cot(2\tan^{-1}x)\}] = 0$

Ans: $\pm 1, \pm(1 \pm \sqrt{2})]$

Sol. $2\cos^{-1}\{\cot(2\tan^{-1}x)\} = n\pi$

$= 0$ if $n = 0$

$= \pi$ if $n = 1$

$= 2\pi$ if $n = 2$

Hence $\cos^{-1}\{\cot(2\tan^{-1}x)\} = 0$ or $\frac{\pi}{2}$ or π

or $\cot(2\tan^{-1}x) = 1, 0,$ or -1 (now proceed by first taking general solution)

Q.9 $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$

Ans. $\frac{\sqrt{3}}{2\sqrt{7}}$

[Sol. $\alpha + \beta = \frac{\pi}{3}$ where $\sin\alpha = x$ and $\sin\beta = 2x$ and $\beta = \frac{\pi}{3} - \alpha$

now $\sin\beta = \sin\left(\frac{\pi}{3} - \alpha\right)$; hence $\sin\beta = \sin\frac{\pi}{3}\cos\alpha - \cos\frac{\pi}{3}\sin\alpha$

$2x = \frac{\sqrt{3}}{2}\sqrt{1-x^2} - \frac{1}{2}x \Rightarrow \frac{5x}{2} = \frac{\sqrt{5}}{2}\sqrt{1-x^2} \Rightarrow 25x^2 = 3(1-x^2)$

$\Rightarrow 28x^2 = 3 \Rightarrow x^2 = \frac{3}{28} \Rightarrow x = \frac{\sqrt{3}}{2\sqrt{7}}$

5. SUMMATION OF SERIES

First change the series into $\tan^{-1}$ form.

Find general term (T_n)

Split the general term in two parts using $\tan^{-1}\left(\frac{a-b}{1+ab}\right) = \tan^{-1}a - \tan^{-1}b$

Apply sigma function

All the terms are cancelled except first and last term.

Q.1 $S_n = \sum_{r=1}^{n} \tan^{-1}\left(\frac{1}{x^2 + (2r-1)x + r^2 - r + 1}\right)$ find S_n & S_∞

or

$S_n = \tan^{-1}\left(\frac{1}{x^2+x+1}\right) + \tan^{-1}\left(\frac{1}{x^2+3x+3}\right) + \tan^{-1}\left(\frac{1}{x^2+5x+7}\right) + \ldots\ldots$n terms find S_n & S_∞

Ans. $S_n = \tan^{-1}(x+n) - \tan^{-1}(x)$; $S_\infty = \frac{\pi}{2} - \tan^{-1}(x)$

Sol. $S_n = \tan^{-1}\left(\frac{(1+x)-x}{1+x(1+x)}\right) + \tan^{-1}\left(\frac{(x+2)-(x+1)}{1+(x+1)(x+2)}\right) + \ldots. + \tan^{-1}\left(\frac{(x+n)-(x+n-1)}{1+(x+n-1)(x+n)}\right)$

$S_n = \tan^{-1}(x+1) - \tan^{-1}x + \tan^{-1}(x+2) - \tan^{-1}(x+1) + \ldots. + \tan^{-1}(x+n) - \tan^{-1}(x+n-1)$

$S_n = \tan^{-1}(x+n) - \tan^{-1}(x)$, $S_\infty = \frac{\pi}{2} - \tan^{-1}x$

Q.2 $S_n = \sum_{r=1}^{n} \tan^{-1}\left(\frac{2r}{r^4 + r^2 + 2}\right)$ find S_n & S_∞

Ans. $S_n = \tan^{-1}(n^2+n+1) - \tan^{-1}(1)$; $S_\infty = \frac{\pi}{4}$

Sol. $S_n = \sum \tan^{-1}\left(\frac{(r^2+r+1)-(r^2-r+1)}{1+(r^2+r+1)(r^2-r+1)}\right)$

$S_n = \tan^{-1}(n^2+n+1) - \tan^{-1}(1)$

$S_\infty = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

Q.3 $S_n = \sum_{r=1}^{n} 4\tan^{-1}\left(\frac{x}{1+r(r+1)x^2}\right)$, $x > 0$ find S_n & S_∞

Ans. $S_n = 4\left[\tan^{-1}(n+1)x - \tan^{-1}(x)\right]$; $S_\infty = 4\left[\frac{\pi}{2} - \tan^{-1}(x)\right]$

Sol. $S_n = 4\sum \tan^{-1}\left(\frac{(rx+x)-rx}{1+rx(rx+x)}\right)$

$Sn = 4[\tan^{-1}(n+1)x - \tan^{-1}x]$

$S_\infty = 4\left[\frac{\pi}{2} - \tan^{-1}x\right]$

Q.4 $S_n = \sum_{r=1}^{n} \tan^{-1}\left(\frac{2}{r^2}\right)$ find S_n & S_∞

Ans. $S_n = \tan^{-1}(n+1) + \tan^{-1}(n) - \tan^{-1}(1)$; $S_\infty = \frac{3\pi}{4}$

Sol. $S_n = \sum \tan^{-1}\left(\frac{2}{1+r^2-1}\right) = \sum \tan^{-1}\left(\frac{(r+1)-(r-1)}{1+(r+1)(r-1)}\right)$

$T_r = \tan^{-1}(r+1) - \tan^{-1}(r-1)$

$T_1 = \tan^{-1}2 - \tan^{-1}0$

$T_2 = \tan^{-1}3 - \tan^{-1}1$

$\vdots$

$T_{n-1} = \tan^{-1}(n) - \tan^{-1}(n-2)$

$T_n = \tan^{-1}(n+1) - \tan^{-1}(n-1)$

$S_n = \tan^{-1}(n+1) + \tan^{-1}(n) - \tan^{-1}(1)$

$$S_\infty = \frac{3\pi}{4}$$

Q.5 $S_n = \sum_{r=1}^{n} \tan^{-1}\left(\frac{1}{2r^2}\right)$ find S_n & S_∞

Ans. $S_n = \tan^{-1}(2n+1) - \tan^{-1}(1)$; $S_\infty = \frac{\pi}{4}$

Sol. $S_n = \sum \tan^{-1}\left(\frac{2}{4r^2}\right) = \sum \tan^{-1}\left(\frac{2}{1+4r^2-1}\right)$

$$S_n = \sum \tan^{-1}\left(\frac{(2r+1)-(2r-1)}{1+(2r+1)(2r-1)}\right)$$

$$= \sum \tan^{-1}(2r+1) - \tan^{-1}(2r-1)$$

$$S_n = \tan^{-1}(2n+1) - \tan^{-1}(1);\ S_\infty = \frac{\pi}{4}$$

Q.6 $S_n = \sum_{r=1}^{n} \tan^{-1}\left(\frac{4r}{r^4-2r^2+2}\right)$ find S_n & S_∞

Ans. $S_n = \tan^{-1}(n+1)^2 + \tan^{-1}(n^2) - \tan^{-1}(1)$; $S_\infty = \frac{3\pi}{4}$

Sol. $S_n = \sum \tan^{-1}\left(\frac{4r}{1+(r^2-1)^2}\right)$

$$S_n = \sum \tan^{-1}\left(\frac{(r+1)^2-(r-1)^2}{1+(r+1)^2(r-1)^2}\right)$$

$$S_n = \tan^{-1}(n+1)^2 + \tan^{-1}(n^2) - \tan^{-1}(1)$$

$$S_\infty = \frac{3\pi}{4}$$